```c
#include <stdio.h>
#include <time.h>

/* O(1) – Constant Time */
int getFirstElement(int arr[], int n)
{
    return arr[0];
}

/* O(n) – Linear Time */
int findMaximum(int arr[], int n)
{
    int max = arr[0];
    for(int i = 1; i < n; i++)
    {
        if(arr[i] > max)
            max = arr[i];
    }
    return max;
}

/* O(n^2) – Quadratic Time */
void bubbleSort(int arr[], int n)
{
    int temp;
    for(int i = 0; i < n - 1; i++)
    {
        for(int j = 0; j < n - i - 1; j++)
        {
            if(arr[j] > arr[j + 1])
            {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

int main()
{
    clock_t start, end;
    double time_taken;
```

```
    int n_values[] = {500, 1000, 2000, 40000};
    int len = sizeof(n_values) / sizeof(n_values[0]);

    printf("Time Complexity Analysis Using Arrays\n");
    printf("------------------------------------\n");

    for(int k = 0; k < len; k++)
    {
        int n = n_values[k];
        int arr[n];

        /* Initializing array */
        for(int i = 0; i < n; i++)
            arr[i] = n - i;

        printf("\nInput Size n = %d\n", n);

        /* O(1) */
        start = clock();
        getFirstElement(arr, n);
        end = clock();
        time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;
        printf("O(1)  Constant Time  : %f\n", time_taken);

        /* O(n) */
        start = clock();
        findMaximum(arr, n);
        end = clock();
        time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;
        printf("O(n)  Linear Time    : %f\n", time_taken);

        /* O(n^2) */
        start = clock();
        bubbleSort(arr, n);
        end = clock();
        time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;
        printf("O(n²) Quadratic Time : %f\n", time_taken);
    }

    return 0;
}
```

Output :

```
Time Complexity Analysis Using Arrays
------------------------------------

Input Size n = 500
O(1)  Constant Time  : 0.000001
O(n)  Linear Time    : 0.000001
O(n²) Quadratic Time : 0.000411

Input Size n = 1000
O(1)  Constant Time  : 0.000001
O(n)  Linear Time    : 0.000002
O(n²) Quadratic Time : 0.001669

Input Size n = 2000
O(1)  Constant Time  : 0.000001
O(n)  Linear Time    : 0.000004
O(n²) Quadratic Time : 0.006313

Input Size n = 40000
O(1)  Constant Time  : 0.000001
O(n)  Linear Time    : 0.000048
O(n²) Quadratic Time : 2.610323
```